

Process Control

Pre-Lab Plan
Unit Operations Lab #1
Date: 1/21/26

Group 3, Tuesday 9 AM
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1. Experimental Objective

The goal of this lab is to optimize different control types (manual, proportional band, and proportional band with integral time) for filling, draining, and maintaining a steady flow within a system. The efficacy of each control method will also be evaluated through a comparison of the discrepancy between the digitally recorded and actual water levels (offset), the stability of the water level over time when disturbances are introduced, and the time required for control changes to be reflected in the system.

2. Experimental

This lab utilizes the Armfield PCT50 Level Control process unit to study the fundamental principles of process control engineering. Initially, the sump tank (Figure 1.1) is filled with deionized water. The PCT50 software is used to operate the process and regulate the transport of fluid from the sump to process tank (1.4). The water level in the process tank is communicated to the control software through an electronic pressure sensor. The inlet flow from the sump to process tank is regulated by the pump (1.2) speed and the inlet valve (1.6.2) and the outflow back into the sump tank is controlled by the outlet valve (1.6.1) and solenoid valve (1.3). The flow through the inlet and outlet valves is determined by the selected threaded orifice (1.5), whereas the pump and solenoid valve allow for more variable control via the control software.

2.1 Apparatus



Figure 1: Apparatus for Process Control Experiment

Table 1: Apparatus for Process Control Experiment

No.	Component	Description
1	Sump Tank	Stores the supply of DI water to be used in the experiments.
2	Centrifugal Pump	To direct DI water into the Process Tank.
3	Solenoid Valve	To allow an extra outflow from the Process tank.
4	Process Tank	Tank in which the level of water will be measured for the experiments.
5	Orifices	2 and 4 mm orifices used in inlet and outlet valve openings
6	Outlet Valve	To control the flow of liquid exiting the Process Tank during the experiment.
7	Inlet Valve	To control the flow of DI water going into the Process Tank.
8	Computer with PCT50 software	Computer that will be used to access Armfield software application

2.2 Materials and Supplies

Table 2: Materials and Supplies for Process Control Experiment

Materials & Supplies	Description
Deionized Water	DI water will be used to fill the sump tank.
2000 mL Beaker	Beaker will be used to fill the sump tank with DI water.

2.3 Experimental Plan

Using the Armfield PCT50 Level Control Process, we can simulate the basic principles of process control engineering. The effects of making changes to the overall system (system or controller configurations) can be shown by making repeatable disturbances to the process. These experiments will show the necessity of proper correct configurations and system characteristics. The Process Control lab consists of three experiments: Experiment 1, where we manually control the water level (open loop), Experiment 2, where we PID control of level in a closed loop, and P control only, and finally Experiment 3, where we PID control of the water level in a closed loop with both P + I control.

For Experiment 1, the main objective is to determine three factors: the pump speed required to maintain a constant level in the process level, the effect of inflow and outflow disturbances on the water level, and the water flow rate when draining and filling the process vessel. By manually changing the pump speed, we can control the level of the water. The independent variables measured in this experiment include the pump speed and opening/closing the solenoid valve. The dependent variables include the measured water levels vs time. With the manipulation of the pump speed and solenoid valve, we track them at these specific parameters:

1. Water level vs time @ 1st pump speed (Initial Level control)
2. Water level vs time @ solenoid on at 1st pump speed and 2nd pump speed (Disturbances)
3. Water level vs time @ solenoid on at 2nd pump speed and 3rd pump speed (Disturbances)
4. Water level vs time during filling (Water flow rate-drain and fill)
5. Water level vs time during filling @ 0% pump speed (Water flow rate-drain and fill)
6. Water level vs time during filling @ 75% pump speed (Water flow rate-drain and fill)

With Experiment 2, the main objective is to determine the effect of different Proportional Band settings (P) on the response (offset or actual level versus setpoint level) and stability of a level process using a P (proportional band) with the only controller that will change being the pump speed. The independent variables in this experiment include the Proportional Band (%), the Setpoint (mm), and the opening/closing state of the solenoid valve. Like Experiment 1, the dependent variables include the measured water levels vs time. At these different configurations of Proportional Band setting, set point, and state of the solenoid valve, the water level vs time will be measured:

1. Water level vs time @ the seven different Proportional Band values (200%, 100%, 50%, 20%, 10%, 5%, and 2%)
2. Water level vs time with an increase outflow disturbance
3. Water level vs time with the change in Setpoint (100mm)

Finally with Experiment 3, the main objective is to determine the effect of different Integral Time settings (I) at a constant P setting on the response (offset or actual level versus setpoint level) and stability of a level process, the difference here we will be using a P + I controller that will change the pump speed. The independent variables in this experiment are the Integral Time setting (I), the same Proportional Band (%) and Setpoint (mm) from Experiment 2, and the opening/closing state of the solenoid valve. The dependent variables will be the water level vs time. We will be measuring the:

1. Water level versus time @ the different Integral Time and constant Proportional Band values
2. Water level versus time with the increase outflow disturbance
3. Water level versus time with the change in Setpoint (100mm)

Project Safety Evaluation (10 pts)

Project Title: Process Control

Date: 1/21/26

	<i>Yes/No</i>	
Potential electrical hazards?	<u>Yes</u>	If yes, identify conditions: Electrical components near water.
Potential mechanical hazards?	<u>No</u>	If yes, identify conditions:
Potential pressure hazards?	<u>No</u>	If yes, identify conditions:
Potential temperature hazards?	<u>No</u>	If yes, identify conditions:
Hazardous/toxic chemicals involved?	No	If yes, hazards involved:
Gloves required?	NO	If yes, specify type:
Other hazards	NO	If yes, specify conditions:
MSDS reviewed by all team members?		List MSDS sheets reviewed: DI Water
<i>Process Parameters</i>	<i>Max Expected</i>	<i>List location and possible hazards and precautions</i>
Temperature (°C)	20 °C	Operating at Room temperature
Pressure (psia)		
<i>Safety Controls</i>	<i>yes/no</i>	<i>If yes, Provide description</i>

I have read relevant background material for the Unit Operations Laboratory entitled: “ **Process Control: Level Control** ” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and

safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

<u>Jazmin Aca</u>	<u>1/20/26</u>
signature of team member	date

<u>Angelica Ramirez</u>	<u>01/20/26</u>
signature of team member	date

<u>Basil Hashim</u>	<u>01/20/26</u>
signature of team member	date

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signature of team member	date